Slide Notes (D.M)

1st slide

Part 1: Definitions :-

- **Proposition** A declarative sentence (a sentence that declares a fact) that is either true or false, but not both.
- **Propositional Variables** Variables that represent propositions, commonly denoted by letters such as p,q,r,s.
- **Propositional Logic** The area of logic that deals with propositions.
- **Compound Propositions** Propositions formed by combining one or more propositions using logical operators.
- **Negation (¬p)** Let p be a proposition. The negation of p, denoted by ¬p, is the statement "It is not the case that p." The truth value of ¬p is the opposite of the truth value of p.
- **Logical Operators (Connectives)** Operators used to form new propositions from two or more existing propositions.
- **Conjunction ($p \land q$)** The proposition "p and q". It is true when **both** p and q are true, and false otherwise.

- **Disjunction ($p \vee q$)** The proposition "p or q". It is false when **both** p and q are false, and true otherwise. (This is the "Inclusive OR").
- **Exclusive OR ($p \oplus q$)** The proposition that is true when **exactly one** of p and q is true, and false otherwise.
- **Conditional Statement ($p \to q$)** The proposition "if p, then q". It is false **only** when p is true and q is false.
    - p is called the *hypothesis* (or antecedent/premise).
    - q is called the *conclusion* (or consequence).
- **Biconditional Statement ($p \leftrightarrow q$)** The proposition "p if and only if q". It is true when p and q have the **same** truth values, and false otherwise. Also called *bi-implications*.
- **Bit String** A sequence of zero or more bits. The length of the string is the number of bits in it.
- **Tautology** A compound proposition that is **always true**, no matter what the truth values of the propositions that occur in it.
- **Contradiction** A compound proposition that is **always false**.
- **Contingency** A compound proposition that is neither a tautology nor a contradiction.
- **Logical Equivalence ($p \equiv q$)** Compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology (i.e., they have the same truth values in all possible cases).

Part 2 : Example and Solution :-

1. Identifying Propositions

- **Problem:** Are the following sentences propositions?
    1. "Toronto is the capital of Canada."
    2. "Read this carefully."
    3. "1+2=3"
    4. "x+1=2"
    5. "What time is it?"
- **Solution:**
    1. **Yes** (False proposition)
    2. **No** (Command)
    3. **Yes** (True proposition)
    4. **No** (Variable x is undefined)
    5. **No** (Question)

2. Negation

- **Problem:** Find the negation of "Today is Friday."
    - **Solution:** "It is not the case that today is Friday" or "Today is not Friday."
- **Problem:** Find the negation of "At least 10 inches of rain fell today in Miami."
    - **Solution:** "It is not the case that at least 10 inches of rain fell..." or "Less than 10 inches of rain fell today in Miami."

3. Conjunction

- **Problem:** Find the conjunction of p: "Today is Friday" and q: "It is raining today."
- **Solution:** "Today is Friday and it is raining today." (True only on rainy Fridays).

4. Translating Conditional Statements

- **Problem:** Let p: "Maria learns discrete mathematics" and q: "Maria will find a good job." Express $p \rightarrow q$ in English.
- **Solution:**
    - "If Maria learns discrete mathematics, then she will find a good job."
    - "Maria will find a good job when she learns discrete mathematics."

- "For Maria to get a good job, it is sufficient for her to learn discrete mathematics."

5. Inverse, Converse, Contrapositive

- **Statement:** "If today is Thursday, then I have a test today."
- **Solution:**
  - **Contrapositive ($\neg q \to \neg p$):** "If I do not have a test today, then today is not Thursday."
  - **Converse ($q \to p$):** "If I have a test today, then today is Thursday."
  - **Inverse ($\neg p \to \neg q$):** "If today is not Thursday, then I do not have a test today."

6. Biconditional

- **Problem:** Let p: "You can take the flight" and q: "You buy a ticket." Express $p \leftrightarrow q$.
- **Solution:** "You can take the flight if and only if you buy a ticket."

7. Translating English to Logic

- **Sentence:** "You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old."
  - **Solution:** Let q = ride, r = under 4ft, s = older than 16. Logic: $(r \wedge \neg s) \to \neg q$.
- **Sentence:** "You can access the Internet from campus only if you are a computer science major or you are not a freshman."
  - **Solution:** Let a = access internet, c = CS major, f = freshman. Logic: $a \to (c \vee \neg f)$.

8. Bitwise Operations

- **Problem:** Find bitwise OR, AND, XOR of `01 1011 0110` and `11 0001 1101`.
- **Solution:**
  - **OR:** `11 1011 1111`
  - **AND:** `01 0001 0100`
  - **XOR:** `10 1010 1011`

9. Proving Logical Equivalence

- **Problem:** Show that $\neg(p \to q)$ and $p \wedge \neg q$ are equivalent.
- **Solution:**
  1. $\neg(p \to q) \equiv \neg(\neg p \vee q)$ (Implication law)
  2. $\equiv \neg(\neg p) \wedge \neg q$ (De Morgan's law)

3. $\equiv p \land \neg q$ (Double negation law)

10. Tautology Proof

- **Problem:** Show that $(p \land q) \to (p \lor q)$ is a tautology.
- **Solution:**
    1. $\equiv \neg(p \land q) \lor (p \lor q)$ (Implication law)
    2. $\equiv (\neg p \lor \neg q) \lor (p \lor q)$ (De Morgan's law)
    3. $\equiv (\neg p \lor p) \lor (\neg q \lor q)$ (Associative/Commutative)
    4. $\equiv T \lor T \equiv T$ (Always True)

11. Practice Exercises

- **Problem 1:** p: "Drive > 65mph", q: "Get ticket".
    - "You will get a ticket if you drive > 65mph" $\to p \to q$.
    - "You get a ticket, but you don't drive > 65mph" $\to q \land \neg p$.
- **Problem 2:** Inclusive vs. Exclusive OR.
    - "Take calculus or computer science": **Inclusive** (You can take both).
    - "Soup or salad": **Exclusive** (You usually pick one).
- **Problem 3:** Truth table for $(p \to q) \land (\neg p \to r)$.
    - (The full truth table is provided in the slide, showing the result column as T,T,F,F,T,F,T,F).

Logical Equivalence Laws :-

- **Identity Laws**
  - $p \wedge T \equiv p$
  - $p \vee F \equiv p$
- **Domination Laws**
  - $p \vee T \equiv T$
  - $p \wedge F \equiv F$
- **Idempotent Laws**
  - $p \vee p \equiv p$
  - $p \wedge p \equiv p$
- **Double Negation Law**
  - $\neg(\neg p) \equiv p$
- **Commutative Laws**
  - $p \vee q \equiv q \vee p$
  - $p \wedge q \equiv q \wedge p$
- **Associative Laws**
  - $(p \vee q) \vee r \equiv p \vee (q \vee r)$
  - $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- **Distributive Laws**
  - $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
  - $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- **De Morgan's Laws**
  - $\neg(p \wedge q) \equiv \neg p \vee \neg q$
  - $\neg(p \vee q) \equiv \neg p \wedge \neg q$

2nd Slide:

Definitions from Predicate Logic Lecture

- **Predicate Logic** An extension of propositional logic that permits concise reasoning about whole classes of entities (statements involving variables).
- **Subject** The variable (e.g., x) in a proposition which denotes the object or entity that the sentence is about.
- **Predicate** The part of a sentence (e.g., "is greater than 3") that denotes a property that the subject can have.
- **Propositional Function** A function P(x) that maps objects to propositions. Once a value is assigned to the variable x, the statement P(x) becomes a proposition with a truth value.
- **Universe of Discourse (Domain)** The collection of all values that a variable x can take. A property is considered true or false relative to this specific domain.
- **Universal Quantification (∀)** The proposition that "P(x) is true for all values of x in the universe of discourse." It is denoted as ∀xP(x).
- **Existential Quantification (∃)** The proposition that "There exists an element x in the universe of discourse such that P(x) is true." It is denoted as ∃xP(x).
- **Scope of a Quantifier** The specific part of a logical expression to which a quantifier applies.

Example and solution of the slide :

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1. Propositional Functions & Truth Values

- **Example 1 (Slide 5):**
  - **Problem:** Let P(x) be "x>3". What are the truth values of P(4) and P(2)?
  - **Solution:**
    - P(4) is "4>3", which is **True**.

        - $P(2)$ is "2>3", which is **False**.
- **Example 2 (Slide 6):**
    - **Problem:** Let $Q(x,y)$ be "$x=y+3$". What are the truth values of $Q(1,2)$ and $Q(3,0)$?
    - **Solution:**
        - $Q(1,2)$ is "1=2+3", which is **False**.
        - $Q(3,0)$ is "3=0+3", which is **True**.
- **Example 3 (Slide 7):**
    - **Problem:** Let $R(x,y,z)$ be "$x+y=z$". What are the truth values of $R(1,2,3)$ and $R(0,0,1)$?
    - **Solution:**
        - $R(1,2,3)$ is "1+2=3", which is **True**.
        - $R(0,0,1)$ is "0+0=1", which is **False**.

2. Universal Quantification (∀)

- **Example (Slide 10):**
    - **Problem:** Let $P(x)$ be "$x+1>x$". Truth value of $\forall x P(x)$ if the domain is all real numbers?
    - **Solution:** Since $P(x)$ is true for all real numbers, $\forall x P(x)$ is **True**.
- **Example (Slide 11):**
    - **Problem:** Let $Q(x)$ be "$x<2$". Truth value of $\forall x Q(x)$ if the domain is all real numbers?
    - **Solution: False**. $x=3$ is a counterexample ($3<2$ is false).

3. Existential Quantification (∃)

- **Example (Slide 12):**
    - **Problem:** Truth value of $\exists x P(x)$ where $P(x)$ is "x2>10" and the domain is positive integers not exceeding 4 {1,2,3,4}.
    - **Solution: True**. The statement expands to $P(1) \vee P(2) \vee P(3) \vee P(4)$. Since $P(4)$ ($16>10$) is true, the entire proposition is true.
- **Example (Slide 13):**
    - **Problem:** Let $P(x)$ be "$x>3$". Truth value of $\exists x P(x)$ for real numbers?
    - **Solution: True**, because there is at least one value (e.g., $x=4$) that makes it true.

4. Nested Quantifiers (Multiple Variables)

- **Example (Slide 15):**
  - **Problem:** Let $Q(x,y,z)$ be "$x+y=z$".
  - **Solution 1:** $\forall x \forall y \exists z Q(x,y,z)$ is **True**. (For every pair x,y, there is a sum z).
  - **Solution 2:** $\exists z \forall x \forall y Q(x,y,z)$ is **False**. (There is no single number z that is the sum of *every* pair x,y) .

5. Practice Problems from Slides 29-31

A. Translating English to Logic (Slide 29)

- **Given:** $P(x)$ = "x can speak Russian", $Q(x)$ = "x knows C++". Domain: Students in your class.
  - "There is a student... who can speak Russian and knows C++" $\Rightarrow \exists x(P(x) \wedge Q(x))$.
  - "There is a student... who can speak Russian but doesn't know C++" $\Rightarrow \exists x(P(x) \wedge \neg Q(x))$.
  - "Every student... either can speak Russian or knows C++" $\Rightarrow \forall x(P(x) \vee Q(x))$.
  - "No student... can speak Russian or knows C++" $\Rightarrow \forall x \neg (P(x) \wedge Q(x))$.

B. Expanding Quantifiers for Finite Domains (Slide 30)

- **Given:** Domain for x,y is {1,2,3}.
  - $\exists x P(x,3) \Leftrightarrow P(1,3) \vee P(2,3) \vee P(3,3)$.
  - $\forall y P(1,y) \Leftrightarrow P(1,1) \wedge P(1,2) \wedge P(1,3)$.
  - $\forall x \forall y P(x,y) \Leftrightarrow$ Conjunction ($\wedge$) of all 9 pairs.
  - $\exists x \exists y P(x,y) \Leftrightarrow$ Disjunction ($\vee$) of all 9 pairs.

C. Truth Values for Integers (Slide 31)

- **Domain:** All integers.
  - $\forall n(n2\geq 0)$ is **True**.
  - $\forall n \exists m(n2<m)$ is **True**.
  - $\exists n \exists m(n+m=4 \land n-m=1)$ is **False** (Solution is non-integer n=2.5).
  - $\forall n \forall m \exists p(p=(m+n)/2)$ is **False** (Sum of integers isn't always even).